CISC 3667: Game Design and Development

Group Project 2: Serious Games - Designing a Game That Makes a Difference

Assigned: Thursday, November 12

Playtest: Thursday, November 19

Due: Tuesday, November 24 **before class**

Groups: 5 students each (one group will have 6 students due to class size)

Project Overview

In this project, your group will design, prototype, playtest, and revise a **Serious Game** (a game created not just for entertainment, but to teach, train, inform, raise awareness, or encourage positive change).

Unlike traditional entertainment games, serious games serve a purpose beyond simply enjoyment. They use gameplay, mechanics, player decisions, and interactions to communicate ideas, develop skills, or encourage players to think about real-world issues.

Examples of serious games discussed in class include The Oregon Trail and Spent.

Your group will choose a real-world topic or issue and design a game that communicates its purpose through gameplay.

You may choose the medium that best supports your intended player experience:

- **Physical Game:** A board game, card game, tabletop game, or another playable physical experience.
- **Digital Game:** A computer, browser-based, or other digital game.

Your choice of medium should be intentional. Consider which format would best communicate your topic, support your mechanics, and create the experience you want players to have.

The goal is not simply to add educational information to a game. The serious purpose should be meaningfully integrated into the gameplay itself.

You will exchange games with another group during the playtest, gather feedback, and make at least one meaningful revision based on your observations.

The goal is to create a game that is both engaging and purposeful, where the gameplay reinforces the intended message, learning objectives, or real-world impact.

Project Schedule

Nov. 12	**Project Assigned:** Form groups, select a topic, and begin designing your game concept.
Nov. 13 - 18	**Design & Prototype Development:** Develop your game concept, mechanics, rules, and a playable prototype.
Nov. 19	**Playtest:** Bring your playable prototype to class and exchange games with another group.
Nov. 20-23	**Revise & Complete:** Use playtest feedback to improve your game and complete your Game Design Document (GDD).
Nov. 24	**Final Submission:** Submit your revised game prototype and completed GDD to Brightspace before class.

Part 1: Design Your Serious Game

Step 1: Choose a Topic and Identify Its Purpose

Begin by identifying a real-world issue, subject, or skill that your game will address.

Your topic should give your game a clear purpose beyond entertainment.

Possible themes include:

- Environmental awareness (climate change, recycling, conservation)
- Public health or safety
- Mental health or mindfulness
- Civic education (voting, social justice, community action)
- Cultural awareness and empathy
- Accessibility and inclusion
- STEM learning (science, mathematics, coding)
- History and heritage exploration
- Financial literacy
- Digital citizenship and online safety

You are encouraged to choose a topic that interests you or that you personally care about.

Your game should have a clearly identifiable serious purpose.

Step 2: Identify Your Target Audience

Determine who your game is designed for.

Consider:

- What age group or demographic is your game intended for?
- What prior knowledge might players have?
- Is the game intended for students, children, adults, professionals, or a general audience?
- What level of difficulty is appropriate?
- Are there accessibility or usability considerations?

Your intended audience should influence your game's mechanics, instructions, presentation, and difficulty.

Step 3: Define the Game Goal and Learning Objectives

Determine what players are trying to accomplish within the game and what they should gain from playing it.

Your game objective should answer:

What are players trying to accomplish?

Your learning or serious objective should answer:

What should players learn, understand, practice, or become more aware of through gameplay?

These objectives should be connected.

For example, a game about environmental conservation might ask players to manage limited resources while balancing the needs of a growing community and protecting the environment.

The player's decisions should **reinforce the subject matter** rather than simply appearing alongside it.

Step 4: Choose Your Medium and Game Format

Decide whether your game will be physical or digital.

Physical Game Examples:

- Board game
- Card game
- Tabletop game
- Cooperative or competitive group game
- Role-playing or simulation activity

Digital Game Examples:

- 2D computer game
- Browser-based game
- Interactive simulation
- Puzzle game
- Narrative-driven game

Consider:

- Which medium best supports your intended experience?
- What types of interactions will players have?
- Does your topic benefit from physical interaction, collaboration, or discussion?

- Would a digital format better support simulation, feedback, progression, or branching decisions?
- What tools and resources can your group realistically use?

You are not required to specifically use Unity or Godot (although you are welcome to if it fits). Choose an appropriate development approach based on your game concept and your group's abilities.

Step 5: Design Your Game Mechanics and Rules

Design the systems that determine how your game works.

Identify:

- What actions can players perform?
- What decisions will players make?
- What challenges or obstacles will they encounter?
- How will players make progress?
- What resources, information, or tools will they use?
- How will player actions affect the game?
- What determines success or failure?
- How does the gameplay reinforce the serious purpose?

Think about the relationship between your mechanics and your intended learning objectives.

For example, if your game addresses financial literacy, players might have to manage a limited budget, make purchasing decisions, and respond to unexpected expenses.

The mechanics should encourage players to engage with the topic through play.

Your rules should clearly explain:

- Setup and instructions
- Player actions and interactions
- Turn structure or gameplay loop
- Challenges and consequences
- Progression and rewards
- Success and failure conditions
- End and winning conditions, if applicable

Avoid mechanics that are unnecessarily complicated or disconnected from the game's purpose.

Step 6: Design the Player Experience

Consider the experience you want players to have.

Ask:

- Is the game enjoyable and engaging?
- Is the objective clear?
- Can players understand what they are supposed to do?
- Is the difficulty appropriate for the intended audience?
- Do players have meaningful choices?
- Does the game encourage critical thinking or problem-solving?
- Is the serious purpose communicated effectively?
- Can players understand the relationship between their actions and the real-world topic?
- What should players think about or take away after playing?

Remember that serious games should balance engagement and educational value.

The game should not feel like a worksheet or lecture disguised as a game. Players should engage with the subject through the gameplay itself.

Step 7: Establish the Game's Visual and Audio Direction

Consider how the presentation supports your game's purpose and intended audience.

Think about:

- Art style and visual theme
- Color palette
- Characters and environments
- Basic User interface (UI), if applicable
- Typography and readability
- Sound effects and music, if applicable
- Tone and atmosphere

Your presentation should be **appropriate for the subject matter**.

For example, a game about environmental conservation might use visual changes in the environment to communicate the consequences of player decisions.

You do not need to produce professional-quality artwork or audio, but your visual and audio choices should support the overall experience.

Step 8: Prepare Your Prototype

Before November 19, create a playable prototype.

Your prototype should be complete enough to demonstrate:

- The game's main objective
- Its serious purpose or learning objective
- The core gameplay loop
- Its primary mechanics and rules
- Player actions and decisions
- How gameplay reinforces the intended purpose
- How players receive feedback
- How the game progresses or ends

For physical games, prepare the necessary components, such as boards, cards, tokens, pieces, and written instructions.

For digital games, prepare a working version that demonstrates the main mechanics and interactions.

Your prototype does not need to be fully polished. However, it must be playable and complete enough for another group to understand and experience the central idea of your game.

Part 2: Playtest and Peer Feedback

On **November 19**, your group will exchange games with another group.

The purpose of the playtest is to evaluate both the gameplay experience and how effectively the game communicates its serious purpose.

Your game should be complete enough that another group can:

- Understand the game's objective and purpose.
- Read the instructions and understand the rules.
- Set up and begin playing.
- Experience the main gameplay mechanics.
- Make meaningful decisions.
- Understand the game's feedback and outcomes.
- Complete the game or experience its intended prototype ending.

The group playing your game should be able to do so without relying on the original designers to explain every mechanic or guide their decisions. Allow the other group to experience the game independently and identify areas that may be confusing or underdeveloped.

What the Playing Group Should Do

After playing another group's game, provide constructive feedback about both the gameplay experience and how effectively the game communicates its serious purpose.

Your feedback should address:

- **Learning Objectives and Serious Purpose:** Was the game's purpose clear? Did the gameplay communicate its intended message or learning objectives?
- **Target Audience and Accessibility:** Was the game appropriate for its intended audience? Were there any accessibility or usability concerns?
- **Engagement and Motivation:** Was the game interesting and engaging? Did you feel motivated to continue playing?
- **Integration of Learning and Gameplay:** Were the learning objectives meaningfully integrated into the mechanics and player decisions?
- **Cognitive Load and Complexity:** Were the rules, instructions, and mechanics easy to understand? Was the game unnecessarily complicated?
- **Feedback, Progress, and Learning Assessment:** Did the game communicate the results of player actions? Could you understand your progress and what you were learning?

- **Real-World Relevance:** Could you connect the gameplay to the real-world topic or issue?
- **Emotional Impact and Tone:** How did the game make you feel? Was its tone appropriate for the subject matter?
- **Accuracy and Content Quality:** Did the information presented seem accurate, clear, and appropriate?
- **Overall Game Experience:** What worked particularly well, and what could be improved?

Provide specific examples whenever possible. Your feedback should help the original designers understand what worked, what did not work, and what could be improved. Be constructive and consider both the strengths of the game and opportunities for revision.

Revise Your Game

After the playtest, review the feedback and identify areas for improvement.

Make **at least one meaningful design revision** based on your observations.

Your revision could involve:

- Clarifying the game's purpose or learning objectives
- Changing or improving a game mechanic
- Revising the rules or instructions
- Adjusting difficulty or progression
- Improving player feedback
- Making learning objectives more integrated into gameplay
- Improving the user interface or physical components
- Addressing accessibility or usability concerns
- Improving the game's pacing or engagement
- Revising content that was confusing or inaccurate

Your final GDD should explain what you observed, what you learned, what you changed, and why you made the change.

Part 3: Game Design Document (GDD) & Final Submission

Group Submission: One GDD and **one playable prototype per group**

Your group will submit a **10-15 page Game Design Document (GDD)** along with your revised game prototype.

The 10-15 page requirement applies to the main body of the GDD. The title page, references, and appendices do not count toward the page total.

The GDD should communicate your game's design vision and document the decisions made throughout development.

Your document should include the following sections:

1. Game Overview

Introduce your game and provide the essential information.

Include:

- Game Title
- Short Description and Tagline
- Genre
- Medium (Digital or Physical)
- Intended Platform or Format
- Target Audience
- Number of Players
- Game Objective
- Serious Purpose / Learning Objectives
- Theme and Setting

2. Game Design and Mechanics

Explain how your game works.

Core Gameplay Loop

- What does the player do repeatedly?
- What actions move the game forward?
- How does the player make progress?

Mechanics and Rules

- Primary player actions
- Challenges and obstacles
- Player decisions and consequences
- Resources, rewards, or penalties
- Progression and difficulty
- Success and failure conditions
- End or winning conditions, if applicable

Components and Interactions

- Physical components or digital systems
- Player interactions
- Controls and input methods, if applicable
- Important game states or values

Include screenshots, diagrams, photographs, or other visuals to help communicate your design.

3. Serious Purpose and Learning Integration

Explain what makes your game meaningful beyond entertainment.

Include:

- What real-world issue, subject, or skill does your game address?
- What should players learn, understand, or become more aware of?
- Why is this topic important to your intended audience?
- How do the mechanics reinforce the learning objectives?
- How do player decisions demonstrate or communicate the subject matter?
- How might players apply what they experience in the game to real-world situations?

Explain why your game is an effective way to communicate its intended message or learning objectives.

Research and Supporting Sources

If your game incorporates factual information about a real-world topic, include **credible sources to support the information presented in your game.** This is particularly important for topics involving science, history, public health, safety, or other factual subjects.

Include **citations in your GDD** and a **reference list at the end of the document.**

Make sure the information presented in your game is accurate, appropriate, and relevant to your intended audience.

4. Player Experience, Presentation, and Accessibility

Describe the experience you want players to have.

Include:

- Intended player experience
- Difficulty and challenge
- Engagement and motivation
- Visual and aesthetic direction
- User interface (if applicable)
- Audio direction (if applicable)
- Accessibility and usability considerations
- How feedback and progression communicate player performance

Explain how these decisions support both the gameplay and the serious purpose.

5. Prototype and Development

Document how your group developed the game.

Include:

- Tools and technologies used
- Development approach
- Important implementation details
- Current prototype features
- Limitations or unfinished features

For **physical games**, describe the materials and components used to create the prototype.

For **digital games**, describe the engine, programming tools, or other technologies used.

Include relevant screenshots, photographs, or diagrams.

6. Playtesting and Iteration

Describe your November 19 playtest and the changes made afterward.

A. Playtest Observations

- What happened when another group played your game?
- Were the instructions and mechanics understandable?
- Did players understand the serious purpose?
- How did players interact with the game?
- What worked particularly well?
- What problems or challenges did players encounter?

B. Feedback Analysis

- What feedback did the other group provide?
- What did the feedback reveal about your game's design?
- Were there any unexpected observations?

C. Revisions

- What did your group change after playtesting?
- Why did you make these changes?
- How did the revisions improve the gameplay or learning experience?

Include examples or screenshots of meaningful changes when appropriate.

7. Team Contributions

Include a list of all group members and their respective roles and contributions.

Examples include:

- Game design and mechanics
- Research and content development
- Programming and implementation
- Visual design and asset creation
- UI/UX design
- Physical component creation
- Documentation and playtesting

Each member should have **identifiable contributions to the project. Multiple students may share responsibilities.** Be specific when describing each member's contributions. Listing a role alone is not sufficient; briefly explain the work each member completed.

8. References

Include a list of all sources used to research and develop your game. This may include articles, books, websites, research papers, educational resources, or other credible materials.

- Cite sources that support factual information presented in your game.
- Include sources used to develop your game's subject matter, learning objectives, or educational content.
- Use a consistent citation format throughout your GDD.
- Make sure all sources cited in the GDD are included in this reference list.

Final Submission Checklist

Submit the following to Brightspace:

- **Final Game Design Document (GDD):** 10-15 pages for the main body, including Sections 1-7. References, the title page, and appendices do not count toward the page total.
- **Revised Playable Prototype:** A digital game, playable link, digital file, physical game, or print-and-play materials, as appropriate.

Only one member from each group is required to submit the final project to Brightspace on behalf of the entire group.

Make sure your prototype is accessible and playable for evaluation. If your game is digital, include the appropriate link or files and any necessary instructions. If your game is physical, include the components and instructions needed to play or evaluate it.

Grading and Group Accountability

All members of a group will receive the same project grade. There is no separate individual reflection submission for this project.

Instead, the GDD must document the group's design decisions, development process, playtesting results, revisions, and team contributions.

All group members are expected to participate meaningfully in the design, development, and completion of the project.

Grading Criteria

Category	Weight	What's Evaluated
Concept & Serious Purpose	15%	Relevance, clarity, and achievability of the serious purpose and learning objectives.
Game Design & Mechanics	20%	Core gameplay loop, mechanics, rules, player choices, engagement, and overall design quality.
Learning Integration & Player Experience	15%	How effectively gameplay reinforces learning objectives and balances engagement with educational value.
Game Design Document (GDD)	20%	Completeness, organization, clarity, professionalism, and documentation of design decisions.
Prototype Functionality	20%	Playability, implementation of core mechanics, and demonstration of the serious purpose.
Playtesting & Iteration	10%	Evidence of testing, analysis of feedback, and meaningful revisions.
Total	**100%**	

Remember: A successful serious game is not simply a game with educational information added.

Its mechanics, decisions, challenges, and player experience should work together to communicate a meaningful purpose.